10.5.3 Gradient and Directional Derivatives

$$\nabla f = \begin{bmatrix} \partial f/\partial x \\ \partial f/\partial y \end{bmatrix}$$

$$D_{\hat{u}} f(x_0, y_0) = \nabla f(x_0, y_0) \cdot \hat{u}, \quad \hat{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, \quad \|\hat{u}\| = 1$$

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \qquad \|\vec{v}\| = \sqrt{v_1^2 + v_2^2} \qquad \hat{u} = \frac{\vec{v}}{\|\vec{v}\|}$$

10.6.1 Maxima & Minima

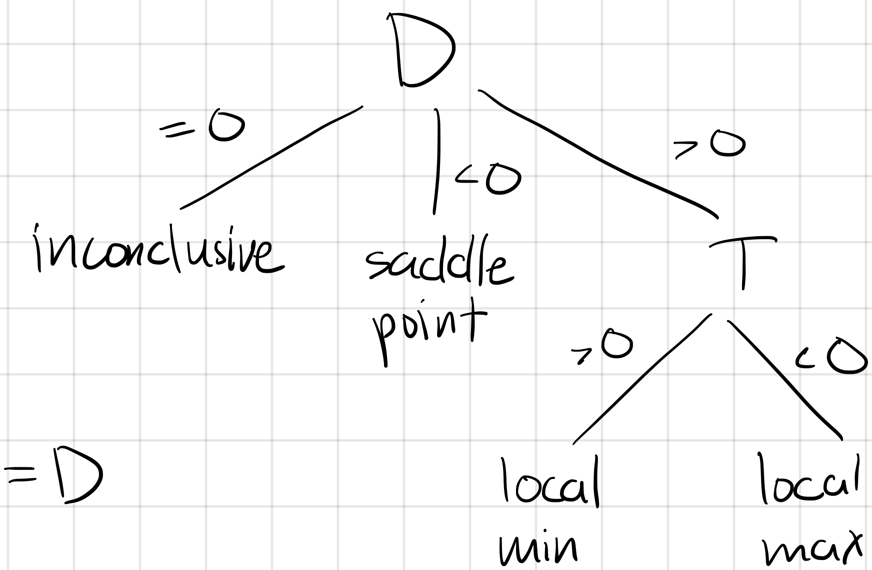
$$f(x, y) \qquad (x_0, y_0)$$

$$\nabla f(x_0, y_0) = \vec{0} = 0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} = (0, 0)$$

$$\text{Hess } f = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix}$$

$$f_{xx} f_{yy} - (f_{xy})^2 = D$$

$$f_{xx} + f_{yy} = T$$



25. $f(x,y) = \sqrt{2x^2 + y^2}$ $\quad (1,2)$ point $\quad \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ vector

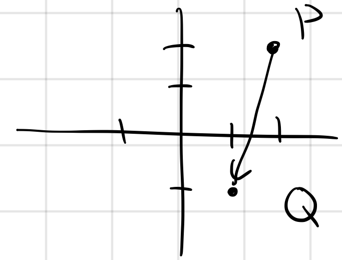
$$\nabla f(x,y) = \begin{bmatrix} \dfrac{2x}{\sqrt{2x^2+y^2}} \\ \dfrac{y}{\sqrt{2x^2+y^2}} \end{bmatrix} \qquad \nabla f(1,2) = \begin{bmatrix} \dfrac{2(1)}{\sqrt{2(1)^2+(2)^2}} \\ \dfrac{2}{\sqrt{2(1)^2+(2)^2}} \end{bmatrix} = \begin{bmatrix} 2/\sqrt{6} \\ 2/\sqrt{6} \end{bmatrix}$$

$$\vec{v} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \qquad \|\vec{v}\| = \sqrt{(1)^2+(1)^2} = \sqrt{2} \qquad \frac{\vec{v}}{\|\vec{v}\|} = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$$

$$\nabla f(1,2)\cdot\vec{u} = \begin{bmatrix} 2/\sqrt{6} \\ 2/\sqrt{6} \end{bmatrix} \cdot \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} = \frac{2}{\sqrt{6}}\cdot\frac{1}{\sqrt{2}} + \frac{2}{\sqrt{6}}\cdot\frac{1}{\sqrt{2}} = \frac{4}{\sqrt{12}} = \boxed{\frac{2}{\sqrt{3}}}$$

scalar!

34. $f(x,y) = e^{x-y}$ $\quad P=(2,2) \quad Q=(1,-1)$



$$\vec{v} = Q - P$$

$$= \begin{bmatrix} 1 \\ -1 \end{bmatrix} - \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} -1 \\ -3 \end{bmatrix}$$

$$\|\vec{v}\| = \sqrt{(-1)^2 + (-3)^2} = \sqrt{10} \qquad \hat{u} = \begin{bmatrix} -1/\sqrt{10} \\ -3/\sqrt{10} \end{bmatrix} \qquad \hat{u} = \frac{\vec{v}}{\|\vec{v}\|}$$

$$\nabla f(x,y) = \begin{bmatrix} e^{x-y} \\ -e^{x-y} \end{bmatrix} \qquad \nabla f(2,2) = \begin{bmatrix} e^0 \\ -e^0 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\nabla f(2,2) \cdot \hat{u} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \cdot \begin{bmatrix} -1/\sqrt{10} \\ -3/\sqrt{10} \end{bmatrix} = 1 \cdot \frac{-1}{\sqrt{10}} + -1 \cdot \frac{-3}{\sqrt{10}} = \boxed{\frac{2}{\sqrt{10}}}$$

5. $f(x,y) = -2x^2 + y^2 - 6y$

$$\nabla f(x,y) = \begin{bmatrix} -4x \\ 2y-6 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \begin{matrix} -4x = 0 & x = 0 \\ 2y - 6 = 0 & y = 3 \end{matrix}$$

$(0,3)$

$$\text{Hess } f(x,y) = \begin{bmatrix} -4 & 0 \\ 0 & 2 \end{bmatrix}$$

$$\text{Hess } f(0,3) = \begin{bmatrix} -4 & 0 \\ 0 & 2 \end{bmatrix}$$

$D = -4 \cdot 2 - 0 \cdot 0 = -8 < 0 \Rightarrow$ saddle (point) @ $(0,3)$

8. $f(x,y) = yxe^{-y}$

$$\nabla f = \begin{bmatrix} ye^{-y} \\ xe^{-y} - xye^{-y} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$ye^{-y} = 0$ ①

$xe^{-y} - xye^{-y} = 0$ ②

① $ye^{y} = 0$

$y = 0$ $\quad \cancel{e^{-y} \neq 0}$

② $xe^{-y} - xye^{-y} = 0$

$xe^{-y}(1-y) = 0$

i) $y = 0$

$xe^{-0}(1-0) = 0$

$x = 0$

$(0, 0)$

$$\begin{bmatrix} 0 & e^{-y} - ye^{-y} \\ e^{-y} - ye^{-y} & -xe^{-y} - (xe^{-y} - xye^{-y}) \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$0 \cdot 0 - 1 \cdot 1 < 0$ saddle point

$(*)$ $f(x)g(x) = 0$

$f(x) = 0 \quad g(x) = 0$

<u>Ex:</u> $(x-1)(x-2) = 0$

$x - 1 = 0 \quad x - 2 = 0$

$x = 1$ or $x = 2$

$$f(x,y) = 2x^2 + 2xy + y^2 + 2x + 3$$

$$\nabla f(x,y) = \begin{bmatrix} 4x + 2y + 2 \\ 2x + 2y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

② $2x + 2y = 0$

$x = -y$

① $4(-y) + 2y + 2 = 0$

$y = 1$

$x = -1$

$$\begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix}$$

$D = (4)(2) - (2)(2) = 4 > 0$

$T = 4 + 2 > 0 \Rightarrow$ local min @ $(-1, 1)$